Sangjun Han

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Research Interests

- Metasurfaces
- Nanophotonics
- Diffractive optics
- Numerical optimization of nanophotonic devices
- Devices using 2-D materials
- Optical measurement, design, simulation, nanofabrication of nanophotonic devices

Education

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
Doctor of Electrical Engineering *Mar. 2020 – Feb. 2025*

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
Master of Electrical Engineering *Feb. 2018 – Feb. 2020*

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
Bachelor of Physics *Mar. 2013 – Feb. 2018*

Research Experiences

Postdoctoral Scholar *Apr. 2025 – Present*
University of California, Berkeley *Berkeley, USA*
Advisor: Hayden Taylor

- Numerical design and fabrication of quartz-based diffractive optical elements for the fast and high-resolution volumetric additive manufacturing
- Investigation of various types of optical beam for the volumetric additive manufacturing

Graduate Research Assistant *Mar. 2018 – Feb. 2025*
Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
Advisor: Min Seok Jang

- Numerical nanophotonic optimization, experimental demonstration, and quasinormal mode analysis of the single-gate electro-optic beam switching metasurfaces
- Development of the active metasurface (metamolecule) for the complex amplitude modulation and its admittance model and equivalent circuit model
- Electromagnetic analysis on the synergistic multimodal gas sensors
- Electromagnetic analysis on the Josephson junction spectroscopy
- Setting-up the automated mid-infrared phase shifting interferometry

Undergraduate Research Assistant *Jan. 2017 – Feb. 2018*
Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea*
Advisor: Min Seok Jang

- Development of the active metasurface (metamolecule) for the complex amplitude modulation

Publications

1. Kim, D., Ahn, D., Kim, H., **Han, S.**, Im, S. G., & Yoo, Y. (2026). Vapor-Phase-Induced Interfacial Interpenetration for High-Performance Polymeric Dehydration Membranes. *Journal of Membrane Science*, 125555. Link

2. Kim, J. Y., Sokhoyan, R., Yi, M., **Han, S.**, Atwater, H. A., & Jang, M. S. (2026). Overcoming Barriers

to Dynamic Phase-only Modulation in Transmissive Metasurfaces via Diffraction Control. *ACS Nano*, 20(8), 6622-6631. Link

3. **Han, S.**, Kong, J., Choi, J., Chegal, W., & Jang, M. S. (2025). Single-gate electro-optic beam switching metasurfaces. *Light: Science & Applications*, 14, 282. Link

4. Fortman, M. A., Harrison, D. C., Rodriguez, R. H., Krebs, Z. J., **Han, S.**, Jang, M. S., ... & Brar, V. W. (2024). Implementing Josephson junction spectroscopy in a scanning tunneling microscope. *arXiv preprint arXiv:2410.03009.* [**Under Review**] Link

5. Kim, Y., Jung, A. W., Kim, S., Octavian, K., Heo, D., Park, C., ... **Han, S.**, Lee, J., Kim, S. & Park, C. Y. (2024). Meent: Differentiable Electromagnetic Simulator for Machine Learning. *arXiv preprint arXiv:2406.12904*. Link

6. Han, H. J., Cho, S. H., **Han, S.**, Jang, J. S., Lee, G. R., Cho, E. N., ... & Jung, Y. S. (2021). Synergistic integration of chemo-resistive and SERS sensing for label-free multiplex gas detection. *Advanced Materials*, 33(44), 2105199. Link

7. **Han, S.**, Kim, S., Kim, S., Low, T., Brar, V. W., & Jang, M. S. (2020). Complete complex amplitude modulation with electronically tunable graphene plasmonic metamolecules. *ACS Nano*, 14(1), 1166-1175. **[Front Cover Article]** Link

International Conferences

1. Kang, J., **Han, S.**, Kim, J., Lee, S., & Jang, M. S., Free-Space Mid-Infrared Complex Modulation by Dual-Channel Graphene Nanoribbon Metasurfaces, *2024 MRS Fall Meeting*, USA. (Dec. 2024)

2. **Han, S.**, Kong, J., & Jang, M. S., Single-gate active beam steering graphene metasurface, *2024 MRS Spring Meeting*, USA. (Apr. 2024)

3. **Han, S.**, Kim, S., Kim, S., Low, T., Brar, V. W., & Jang, M. S., Electrically Tunable Metasurface for Complex Amplitude Modulation, *META 2021 The 11th International Conference on Metamaterials, Photonic Crystals and Plasmonics*, Poland. (Jul. 2021)

4. **Han, S.**, Cha, J., Kim, S., Kim, S., Low, T., Brar, V. W., & Jang, M. S., Complete complex amplitude modulation with graphene plasmonic active metamolecules, *SPIE Optics + Photonics 2020*, Online (USA). (Aug. 2020)

5. **Han, S.**, Kim, J., Kim, J. Y., Park, J., & Jang, M. S., Electronical Tuning of Complex Reflectivity with Graphene-based Metasurface, *2019 MRS Fall Meeting & Exhibit*, USA. (Dec. 2019)

6. **Han, S.**, Kim, S., Menabde, S., Kim, S., Low, T., Brar, V. W., Atwater, H., & Jang, M. S., Recent Advances in Mid-Infrared Graphene Plasmonics: Metasurface for Complex Amplitude Modulation and Compact Waveguide Switch, *RPGR 2019: Recent Progress in Graphene and 2D Materials Research*, Japan. (Oct. 2019)

7. **Han, S.**, Kim, S., Menabde, S., Brar, V. W., Atwater, H., & Jang, M. S., Recent Advances in Mid-Infrared Graphene Plasmonics: Metasurface for Complex Amplitude Modulation and Compact Waveguide Switch, *META 2019 The 10th International Conference on Metamaterials, Photonic Crystals and Plasmonics*, Portugal. (Jul. 2019)

8. **Han, S.**, Ha, H., & Jang, M. S., Electrical tuning of complex reflectivity with graphene-based metasurface, *The 9th International Conference on Surface Plasmon Photonics*, Denmark. (May. 2019)

9. Park, J., Kim, S., **Han, S.**, Ha, H., Menabde, S., & Jang, M. S., Ultimate light trapping in the free-form MIM plasmonic waveguide, *The 9th International Conference on Surface Plasmon Photonics*, Denmark. (May. 2019)

Patents

1. **Chinese Patent:** CN 111856783B Link
   Title: 光器件及其制作方法
   Date: March 14, 2025
   Inventors: Min Seok Jang, **Sangjun Han**, Shinho Kim

2. **Korean Patent:** KR 102730614B1 Link
   Title: 광소자 및 그의 제조방법
   Date: Novemeber 12, 2024
   Inventors: Min Seok Jang, **Sangjun Han**, Shinho Kim

3. **US Patent:** US 11747706B2 Link
   Title: Optical display device having variable conductivity patterns
   Date: September 5, 2023
   Inventors: Min Seok Jang, **Sangjun Han**, Shinho Kim

4. **PCT Patent:** WO 2020218698A1 Link
   Title: 광소자 및 그의 제조방법
   Date: October 29, 2020
   Inventors: Min Seok Jang, **Sangjun Han**, Shinho Kim

Awards

1. **Outstanding Ph.D. Thesis Award**
   Date: March 31, 2025
   Institution: School of Electrical Engineering, KAIST, South Korea

2. **Program Directors Award**
   Date: April 21, 2023
   Institution: Samsung Science & Technology Foundation, South Korea

3. **Kim Choong-Ki Award: Best Research Achievement Award**
   Date: April 5, 2021
   Institution: School of Electrical Engineering, KAIST, South Korea

Technical Expertise

Device fabrication

- Graphene transfer
- Atomic layer deposition (ALD)
- Metal thin-film depostion
- Photolithography (DUV, Maskless)
- Quartz wafer based fabrication
- Wafer membrane handling, Wafer dicing
- Wire bonding

Automated Set-up for Optical Measurements

- Phase shift interferometry
- Beam steering measurement set-ups

  Including: Lock-in amplifier, Detectors (MCT, Thermocouple, Mid-IR array detector), Quantum cascade laser (QCL), Keithley sourcemeter

Characterization Equipment

- Fourier transfrom infrared spectroscopy (FTIR)
- Atomic force microscopy (AFM)
- Scattering Scanning Near-field Optical Microscopy (s-SNOM)

Simulation Software/Progamming Skills

- Lumerical FDTD (Advanced)
- COMSOL Multiphysics (Advanced; Wave optics, AC/DC)
- Rigorous coupled wave analysis (RCWA, Advanced; Reticolo, S4)
- Numerical optimization algorithms (Advanced; Genetic algorithms, BOBYQA)
- Fourier (or Hankel)-based electromagnetic wave propagation simulation
- MATLAB (Advanced; Automated optical set-up, Post-processes for the experiments, RCWA, Optimization algorithms, Quasinormal mode analysis)
- Python (Intermediate; RCWA, Optimization algorithms), Wolfram Mathematica (Intermediate)
- AutoCAD (Intermediate), KLayout (Intermediate)